

PROJECT REPORT OF CO-OP Project at Industry (Module-2)
ON
Impact of Social Media on Adolescent Mental Health
submitted in partial fulfilment of the requirements for the award of degree of
BACHELOR OF ENGINEERING
In
COMPUTER SCIENCE AND ENGINEERING

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DECLARATION

I hereby certify that the work which is being presented in the project report entitled “**Impact of Social Media on Adolescent Mental Health**” in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of Dr. Darpan Anand. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

Place: Rajpura

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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ABSTRACT

This research project investigates the impact of social media usage on adolescent mental health using Artificial Intelligence and Machine Learning techniques.

With the rapid growth of social media platforms, concerns regarding mental health issues such as anxiety, depression, and stress among adolescents have significantly increased.

The study utilizes machine learning models such as Linear Regression and Random Forest to analyze patterns and correlations between social media usage and mental health indicators.

A dataset consisting of behavioral, emotional, and usage-related attributes is analyzed to identify key factors affecting mental well-being.

The results highlight meaningful relationships between excessive social media usage and negative mental health outcomes.

The project demonstrates how data-driven approaches can be used to understand psychological trends and support early intervention strategies.

1. INTRODUCTION

In recent years, social media has become an integral part of adolescents' daily lives. Platforms such as Instagram, Facebook, and Twitter have transformed communication, entertainment, and information sharing. However, excessive use of these platforms has raised concerns regarding their impact on mental health.

This project aims to analyze how social media influences adolescent mental health using AI/ML techniques. By leveraging data analysis and predictive modeling, the study attempts to uncover patterns that contribute to mental health issues.

1.1 Problem Statement

Adolescents are increasingly exposed to social media, which may lead to mental health challenges such as anxiety, depression, low self-esteem, and social isolation. Traditional analysis methods are insufficient to uncover complex patterns in such data.

1.2 Objectives

1. To analyze the relationship between social media usage and mental health
2. To apply machine learning models for predictive analysis
3. To identify key factors affecting adolescent mental well-being
4. To evaluate model performance and interpret results.

2. METHODOLOGY

The project follows a structured data science and machine learning approach:

2.1 Data Collection

Data was collected from publicly available datasets and surveys focusing on adolescent behavior, screen time, emotional state, and social interactions.

2.2 Data Preprocessing

- Handling missing values
- Data normalization and encoding
- Feature selection

2.3 Model Selection

Two machine learning models were used:

- **Linear Regression** – for identifying relationships between variables
- **Random Forest** – for handling complex patterns and improving prediction accuracy

2.4 Model Training and Testing

The dataset was split into training and testing sets to evaluate model performance.

2.5 Evaluation Metrics

- Accuracy
- Mean Squared Error (MSE)
- R-squared value

3. TOOLS AND TECHNOLOGIES

The following tools and technologies were used:

- **Programming Language:** Python
- **Libraries:** NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn
- **Machine Learning Models:** Linear Regression, Random Forest
- **Development Environment:** Jupyter Notebook / Google Colab
- **Data Visualization:** Matplotlib, Seaborn

These tools enabled efficient data analysis, visualization, and model implementation.

4. IMPLEMENTATION

The implementation phase involved applying machine learning techniques to analyze the dataset.

4.1 Data Analysis

Exploratory Data Analysis (EDA) was performed to understand trends, correlations, and distributions in the dataset.

4.2 Feature Engineering

Important features such as screen time, number of social media platforms used, sleep duration, and emotional indicators were extracted.

4.3 Model Implementation

- Linear Regression was used to establish relationships between variables
- Random Forest was used for better prediction and handling non-linear data

4.4 Visualization

Graphs and plots were generated to represent:

- Correlation between social media usage and mental health
- Feature importance
- Prediction results

5. MAJOR FINDINGS / RESULTS

The study revealed several important findings:

- Increased social media usage is strongly associated with higher levels of anxiety and depression
- Lack of sleep due to excessive screen time negatively affects mental health
- Random Forest model performed better than Linear Regression in terms of accuracy
- Certain features such as **screen time and social comparison** had a significant impact on mental health outcomes

The results indicate that machine learning can effectively identify patterns in behavioral data.

6. CONCLUSION AND FUTURE SCOPE

Conclusion

This project successfully demonstrates the application of machine learning techniques in analyzing the impact of social media on adolescent mental health. The findings highlight the importance of monitoring social media usage and promoting healthy digital habits.

Future Scope

- Use deep learning models for more accurate predictions
- Incorporate real-time data collection from apps
- Develop a mental health prediction system or app
- Extend the study to different age groups and demographics.

REFERENCES

- Research papers on adolescent mental health
- Scikit-learn official documentation
- Kaggle datasets
- www.geeksforgeeks.org
- www.w3schools.com

APPENDICES

- Dataset description
- Python implementation code
- Graphs and visualizations
- Model output samples